

COURSE TITLE : FABRIC FORMATION III PRACTICAL
COURSE CODE : 5067
COURSE CATEGORY : A
PERIODS/WEEK : 4
PERIODS/SEMESTER : 52
CREDITS : 2

COURSE OUTCOME

Sub	Student will be able to
1	Operate Jacquard looms
2	Development of Jacquard designs
3	Practice harness mounting, card cutting & lacing
4	Set mechanism for automatic, multiple box looms & Rapier loom
5	Estimate the costing of woven fabrics
6	Practice Textile CAD

LIST OF EXPERIMENTS

1.0 JACQUARDS

- 1.1 Develop a full Jacquard design from a motif figure and prepare the same for card cutting
- 1.2 Perform the card punching, card lacing and weave the design in single lift Jacquard
- 1.3 Study single lift single cylinder Jacquard
- 1.4 Identify various parts of single lift single cylinder Jacquard
- 1.5 Find the count of comber board
- 1.6 Identify the various parts of a Jacquard machine
- 1.7 Understand the working of double lift double cylinder jacquard
- 1.8 Analyze different Jacquard sample and prepare the details required to reproduce the same (10 - Samples-minimum)

2.0 AUTOMATIC LOOMS, MULTIPLE BOX LOOMS AND RAPIER LOOMS

- 2.1 Dismantle and assemble the weft feeler mechanism
- 2.2 Set the cop changing mechanism
- 2.3 Set the warp stop motion
- 2.4 Set the let off motion
- 2.5 Set the weft cutters
- 2.6 Operate the automatic loom
- 2.7 Study of drop box motion available in the laboratory
- 2.8 Arrange the pattern card for drop box loom for various weft colouring schemes
- 2.9 Set the drop box as per timing
- 2.10 Study positive let- off motion
- 2.11 Set single thread warp stop motion
- 2.12 Understand the working principle of Rapier picking mechanism

3.0 ANALYSIS AND COSTING OF FABRIC

- 3.1 Understand the factors influencing costing
- 3.2 Determine the quantity and of warp and weft required to produce 100 meters of cloth
- 3.3 Method of estimating Ex-mill cost of one meter cloth
- 3.4 Estimate the Ex-mill cost one meter of grey fabric
- 3.5 Estimate the Ex-mill cost one meter of bleached fabric
- 3.6 Estimate the Ex-mill cost one meter of dyed fabric
- 3.7 Estimate the Ex-mill cost one meter of check fabric
- 3.8 Identify the cost of production by analyzing sample cloths

4.0 COMPUTER AIDED DESIGNING

- 4.1 Computer aided woven design software.
- 4.2 Study of colour library
- 4.3 Creation of yarn library
- 4.4 Study of software for dobby design
- 4.5 Study of software for Jacquard design
- 4.6 Creation of Fabric simulation

LIST OF MACHINERIES & EQUIPMENTS

- 1. Handlooms mounted with 100 hooks Jacquard
- 2. Handlooms mounted with 200 hooks Jacquard
- 3. Handlooms mounted with Cross border Jacquard (200 hooks)
- 4. Power loom (36'') with Double lift double cylinder Jacquard
- 5. Drop box loom 36''- 1x4 boxes with card saving mechanism
- 6. Automatic cop changing loom 48''/52''
- 7. Projectile looms, with four colour weft insertion-180/190cms
- 8. Rapier looms with four colour weft insertion.
- 9. Dobby shedding with +ve let-off motion with reed space 190cm, four colour weft insertion
- 10. Computer aided design for Dobby, Jacquard and cloth simulation, woven pattern etc.
- 11. Computer system for CAD applications.
- 12. Appropriate Tools